

Project - Network Redesign

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1 Logical Network Redesign

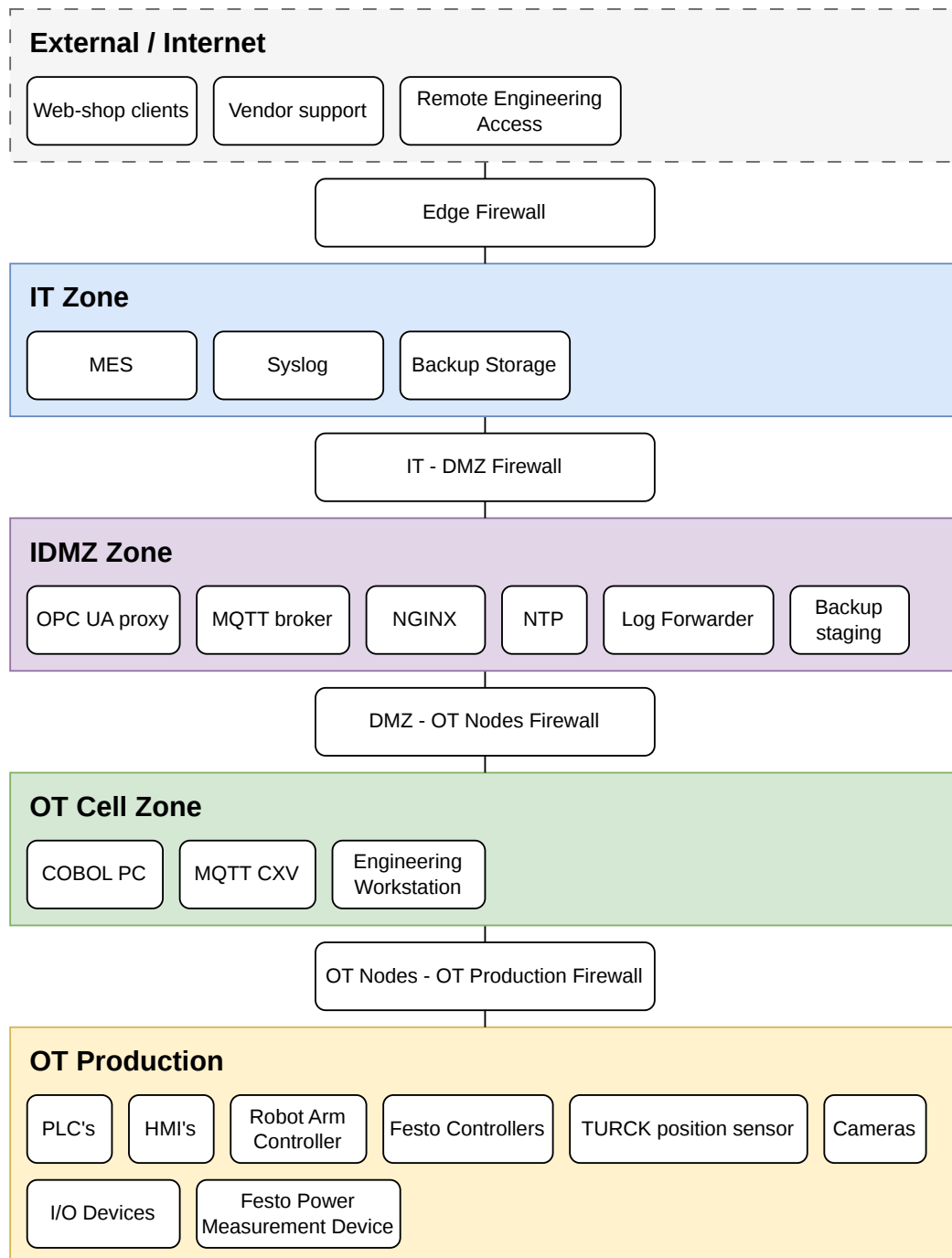


Figure 1: Zone & Conduit Diagram

The current flat /18 network is replaced with a layered zone architecture based on the IEC 62443 zone and conduit model. Each zone groups devices with similar operational functions. All inter-zone traffic passes through a firewall and is restricted to explicitly whitelisted traffic paths.

1.1 Logic Zones

1.1.1 IT Zone

The IT Zone contains nodes handling business logic and order management. It have no direct routed path to any OT device - all IT to OT traffic goes through the IDMZ.

1.1.2 IDMZ

The IDMZ acts as a buffer between IT and OT devices. No connection is allowed to pass from the IT to OT and vice versa. Every traffic exchange terminates inside IDMZ, and its re-initiated to the other side. This solution implements IEC 62443-3-3 SR 5.1 principle.

1.1.3 OT Cell Zone

The OT Cell Zone Contains devices supporting production, but not being part of the physical production process. The Engineering Workstation is placed in this logical zone, however, in physical implementation, this node will be further isolated into separate VLAN.

1.1.4 OT Production Zone

The OT Production Zone contains devices being part of the direct physical production process. Only inbound traffic allowed in this zone are reads and writes from the proxy services, and S7comm+ from the engineering workstation.

1.2 New Nodes

1.2.1 OPC UA proxy

In the AS-IS architecture, MES can directly connect to the PLC's OPC UA servers over the flat network. OPC UA proxy node aggregates all PLC's OPC UA servers traffic into single endpoint inside IDMZ. The MES connects only to proxy, that is also responsible for enforcing rules on what nodes are readable and writable from the MES. Furthermore, it logs all sessions. This solution satisfies IEC 62443-3-3 SR 3.2.

1.2.2 Nginx

In AS-IS architecture, MES can directly access unencrypted HTTP services from the OT nodes. As stated in security requirements, all not strictly required HTTP servers of OT devices will be disabled. However, for required HTTP traffic (e.g PLC's alarm table j- MES) the Nginx node is introduced in IDMZ. MES connects to the Nginx over HTTPS TLS 1.2, terminates the connection, access the requested HTTP node server and forwards it to the MES. Nginx enforces authentication, logging all requests and block requests for services that should not be reached from IT zone.

1.2.3 Dedicated MQTT broker

In the AS-IS architecture MQTT CVX device publish messages over a flat network to the MES. MQTT broker node placed in IDMZ, aggregates MQTT traffic from the CVX device using TLS 1.2 encryption, and it restrict inbound traffic to publish messages only. MES is subscribes to the MQTT

broker, preventing direct communication between OT node and IT node. This solution satisfies IEC 62443-3-3 SR 1.1 and SR 3.1.

1.2.4 Engineering Workstation

In the AS-IS architecture, TIA Portal is run on the same PC as MES service - part of IT and reachable over flat network. A dedicated Engineering Workstation is placed in the OT Cell Zone (physically on separate VLAN). This workstation purpose is to program and configure PLC's and HMI's with separation from IT nodes. Traffic is restricted only to TCP 102 on PLC's and HMI's. This enforces least-privilege and IEC 62443-2-1 §4.3.3.

1.2.5 Syslog

In the AS-IS architecture, there is no centralized logging system implemented. Syslog node collects all logs from the firewalls, managed switches, OPC UA and HTTP proxy, and MQTT broker. This provides post-attack analysis and satisfies IEC 62443-3-3 SR 6.1 and NIS2 Article 21(2)(f).

1.2.6 Backup

In AS-IS architecture, there is no documented backup procedure. The backup node receives configuration backups from the Engineering Workstation after every change and daily database backups from the MES. It is placed on separate VLAN inside IT zone, separated from inbound traffic from OT. Satisfies IEC 62443-2-1 §4.3.4 and ISO/IEC 27001 A.12.3.

1.2.7 Log & Backup proxy

1.3 Firewalls

This section only describes briefly the high-level purpose of the firewall nodes. Further specification on permitted conduits is described in 7.2.2.

1.3.1 Edge Firewall

Terminates all VPN tunnels from remote engineers and vendor support, enforces MFA before any traffic enters the IT zone, and blocks all unsolicited inbound connections from the internet. Default-deny on all inbound. Satisfies IEC 62443-2-1 §4.3.3.6 and NIS2 Article 21(2)(i).

1.3.2 IT - DMZ Firewall

Controls all traffic between the IT zone and the IDMZ.

1.3.3 DMZ - OT Cell Firewall

Controls traffic between the IDMZ and the OT cell zone.

1.3.4 OT Cell - OT Production Firewall

Controls traffic between the OT cell zone and the six station VLANs. OPC UA proxy connections arrive from the IDMZ side, not the cell zone side, so they do not pass this firewall. Profinet never crosses this boundary — it remains Layer 2 within each station. All other traffic denied. Satisfies IEC 62443-3-3 SR 5.1 and the least privilege principle.

2 Network Implementation

The Network Implementation diagram is appended to the report for better visibility.

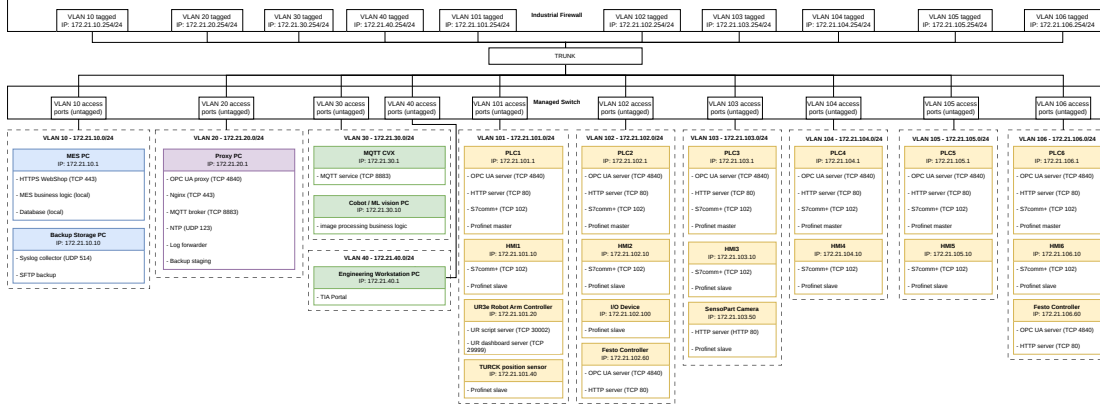


Figure 2: Network Implementation

2.1 VLAN Segmentation & Restricted Routing

The redesigned factory network is implemented using VLAN-based segmentation on managed industrial switches. Each security zone is assigned a dedicated VLAN and IP subnet. Inter-VLAN communication is not performed by the switch itself but is routed through a central industrial firewall. The firewall has one tagged VLAN port for each VLAN - which is a default gateway for each node within that VLAN. The firewall enforces security policies between zones. A VLAN trunk connection between the firewall and the managed switch transports traffic for all security zones while preserving logical separation through IEEE 802.1Q VLAN tagging.

Because the switch never bridges VLANs, the only path between any two zones is through the firewall. The firewall applies a default-deny policy: traffic between VLANs is dropped unless a rule explicitly permits it. Traffic within a VLAN (for example PLC-to-HMI Profinet inside one station) is switched locally and never reaches the firewall. Profinet RT traffic is Layer 2 only and is confined to its station VLAN; it is never routed across the firewall. This realises the four logical firewall boundaries (Edge, IT-IDMZ, IDMZ-OT Cell, OT Cell-OT Production) on a single physical firewall, where each boundary corresponds to a rule set between the relevant VLAN interfaces. It satisfies IEC 62443-3-3 SR 5.1 (Restricted Data Flow) and SR 5.2 (Zone Boundary Protection).

VLAN	Zone	Nodes
10	IT	MES PC, Data Storage PC
20	IDMZ	Proxy PC
30	OT Cell	COBOL PC, MQTT CVX
40	OT Cell - Engineering	Engineering Workstation
101	OT Prod - Station 1	PLC1, HMI1, UR3e Robot Arm Controller, TURCK position sensor
102	OT Prod - Station 2	PLC2, HMI2, I/O device, Festo Controller
103	OT Prod - Station 3	PLC3, HMI3, SensoPart Camera
104	OT Prod - Station 4	PLC4, HMI4
105	OT Prod - Station 5	PLC5, HMI5
106	OT Prod - Station 6	PLC6, HMI6, Festo controller

2.2 Justified Protocols Routes

The permitted flows exists to allow a required operational functionality - all other are denied. The IT node never reaches an OT node directly. All IT to OT connections are terminated and re-initiated in the IDMZ proxy to the other side.

Flow	Path	Protocol/Port	Justification
Process Data, MES to PLC	MES to OPC UA proxy to PLC's OPC UA server	OPC UA/TCP4840	MES reads/writes process data. MES connection is terminated in OPCUA proxy. No direct connection between MES and PLC.
Web data, MES to OT nodes	MES to Nginx to OT nodes web server	HTTPS/TCP443 (IT side), HTTP/TCP80 (OT side)	Required Web data from OT nodes is fetched by Nginx reverse proxy instead of MES directly - plain text HTTP stays in OT zone, and Nginx connects to MES through HTTPS.
Telemetry, MQTT CVX device to MES	MQTT CVX to MQTT broker to MES (subscribe)	MQTT/TCP8883 TLS	OT only publish, IT only subscribe
PLC/HMI programming	Engineering Workstation to HMI's/PLC's	S7comm+/TCP102	Programming only from the dedicated workstation in the engineering VLAN
Cobot process data	Cobot/MLvisionPC to PLC2	OPC UA/TCP4840	Vision PC needs PLC2 data only; scoped to Station 2
Time synchronization	All devices to NTP server	NTP/UDP123	Single time source for all OT devices
Logging	All devices to Syslog collector	Syslog/UDP514	Centralised log collection for monitoring
Webshop	Internet clients to MES	HTTPS/TCP443	Public webshop terminates in IT, no OT reachability